PHYSICAL ACTIVITY AND SELF-REPORTED MEMORY DECLINE AMONG U.S. ADULTS AGED 50 YEARS AND OLDER:
A SEX-STRATIFIED ANALYSIS OF THE 2023 BEHAVIORAL RISK FACTOR SURVEILLANCE SYSTEM(BRFSS)

A Master's thesis submitted in partial fulfillment of the requirements for the degree of Master of Science

Department of Biostatistics

School of Global Public Health, New York University

May 2026

ABSTRACT

Introduction

Self-reported memory decline is recognized as a clinically relevant early indicator of subsequent cognitive impairment and dementia, and physical activity is a potentially modifiable behavior associated with better cognitive health. Recent nationally representative U.S. evidence on the association between physical activity and self-reported memory decline remains limited, particularly regarding whether this relationship differs by sex. As the prevalence and incidence of dementia vary by sex, understanding the effectiveness of specific strategies for males and females separately helps define more targeted prevention strategies for dementia. Therefore, this study examined the association and potential effect modification by sex among U.S. adults aged 50 years and older.

Methods

We performed a cross-sectional analysis of the 2023 Behavioral Risk Factor Surveillance System (BRFSS) restricted to U.S. adults aged 50 years and older. After excluding respondents with missing data on key variables, 35,366 records were retained. Survey-weighted logistic regression models were used to evaluate the association between physical activity and self-reported memory decline, including a crude model, an adjusted model, and a sex-interaction model. Sex-stratified models were additionally fit for males and females.

Results

Higher physical activity was associated with lower odds of self-reported memory decline in the crude model, with a graded inverse pattern across activity levels. After adjustment for sex, age group, education, income, smoking, and depressive disorder, only the highly active group remained significantly associated with lower odds of memory decline (adjusted OR=0.76). In sex-stratified models, this inverse association remained evident in both females (OR=0.73, p=0.001) and males (OR=0.80, p=0.02). However, interaction terms between physical activity and sex were not statistically significant, suggesting no clear sex difference in the association.

Conclusion

Higher physical activity was associated with lower odds of self-reported memory decline among U.S. adults aged 50 years and older, with no clear evidence that this association differed by sex.

These findings support physical activity promotion as a population-level strategy for preserving cognitive health and informing dementia prevention efforts in older adults.

Key Words: Physical Activity; Self-Reported Memory Decline; Dementia Risk; Older Adults

Abstract Word Count: 318

Main Text Word Count: 3,812

Number of Tables: 4

Introduction

Prevalence, trend, and global impact of dementia

Dementia has become one of the most prevalent chronic conditions among older adults worldwide. Recent estimates show substantial cross-country variation in prevalence. In China, the age- and sex-adjusted prevalence of dementia was 6.0% among adults aged ≥60 years, representing approximately 15.0 million individuals[1]. In the United States, the prevalence of Alzheimer's disease dementia was estimated at 8.5% among adults aged ≥65 years[2]. In India, dementia prevalence was 7.4% among adults aged ≥60 years, or about 8.8 million individuals nationally[3]. In Mexico City, prevalence reached 14% among older adults[4]. Together, these findings underscore the substantial global burden of dementia and its major social, economic, and quality-of-life consequences for affected individuals and their caregivers.

Subjective memory complaints as risk indicators of dementia

While objective cognitive testing remains the diagnostic standard for dementia, growing evidence suggests that subjective memory complaints (SMCs) may be important early markers of elevated dementia risk. Previous studies have reported two notable findings. First, individuals with increasing memory complaints over time tend to show steeper cognitive decline and greater subsequent dementia risk, even after accounting for mood and demographic factors[5,6]. Second, subjective memory complaints have been shown to predict incident dementia in cognitively intact older adults, but not in older adults with existing cognitive impairment[7]. These findings suggest that perceived memory decline is a meaningful early risk marker for dementia, particularly when self-reports are supported by informant reports, given concerns about the reliability of self-assessment in some individuals with impaired cognition. Clinically, this

highlights the value of recognizing self-reported memory decline as a potential early signal for timely monitoring and prevention in public health and primary care settings.

Physical activity as a modifiable approach for cognitive health

There have been solutions to relieve dementia and improve cognition, including pharmacological approaches such as the utilization of statins[8], or the prevention of delirium[9], suggesting potential preventive benefits clinically. On the Behavioral side, approaches to manage modifiable risk factors such as hypertension, obesity, diabetes, depression, inactivity, alcohol, and smoking provide additional solutions to dementia[10,11].

Physical activity is one of the accessible behavioral approaches related to cognitive health. Evidence from trials and reviews suggests that structured physical activity improves outcomes relevant to older adults, including global cognition, executive function, memory, mobility, and psychological well-being. For example, multicomponent interventions that combine aerobic and strength training have shown reliable benefits among people living with dementia[12-14]. In addition, programs that pair movement with engaging formats, such as interactive or game-based activities, appear feasible and may enhance psychomotor speed and adherence[15].

Given that physical activity has been consistently linked to reduced dementia risk and that self-reported memory decline is recognized as an early marker of dementia, it is therefore relevant to examine whether physical activity is also associated with self-reported memory decline. Investigating this association may provide valuable insight into modifiable behavioral factors that support cognitive health and help guide public health strategies focused on prevention and early intervention before clinically apparent impairment develops.

Sex-specific difference on prevalence and incidence of dementia

The prevalence of dementia among females is significantly higher than that among males in the United States, which has been suggested in some papers to be due to women generally living longer than men. [16]While the sex difference in the incidence of dementia does not have a unified view, some studies in the United States found that there is no significant sex difference in the incidence of dementia[17,18], while others suggest that women have a higher incidence of dementia than men.[19,20]

These inconsistencies across studies might be due to the use of different diagnostic criteria for dementia, sample sizes, or age ranges across study populations. Notably, after an AD diagnosis, women tend to experience faster cognitive decline and disease progression.[21,22]

The causes of sex differences in the prevalence and incidence of dementia have been observed and investigated from various aspects. In addition to biological sex differences (such as head size, brain volume, or gonadal hormones), sociodemographic factors such as education, work, and social expectations received by men and women, which therefore affect their cognitive health and health behaviors, are also considerable.[23-26] For example, people with higher cognitive reserve (such as those with higher education, a stronger occupational history/skilled work, and a better diet) are found to be more resilient to pathological insults to the brain.[27,28] Through improved individual baseline cognitive performance, it takes longer for them to reach the dementia threshold.[29] In the last century, men had a greater chance of getting a better job or receiving a better education than women, corresponding to those older people currently being tested for dementia. However, as opportunities for women to access higher education and skilled employment have expanded substantially in recent decades, women now attain similar average levels of education than men in the United States.[30] These generational shifts in the educational and occupational profiles of women may influence the prevalence and incidence of dementia across sexes over time. Similarly, physical activity participation is also shaped by gender roles in society, as caregiving and household responsibilities have been shown to disproportionately reduce women's engagement in leisure time physical activity, particularly during and after parenthood,[31] though recent decades have seen modest shifts in the gendered distribution of caregiving and household labor. [32] Nationally representative data from the United States have consistently shown that women are less likely than men to meet recommended physical activity levels, with this gap persisting among adults aged 50 years and older.[33] As physical activity is identified as an approach to support cognitive health, these sex based differences in activity patterns have implications for the prevalence and incidence of dementia. Given these sex differences in dementia outcomes and the social and behavioral pathways that contribute to them, examining whether the protective association of physical activity with self-reported memory decline differs by sex can inform more targeted prevention strategies.

Literature Conclusions

Much of the prior literature has been conducted in European cohorts, selected clinical samples, or relatively small study populations[34-36], which may limit the generalizability of findings to the broader U.S. population. Existing U.S. studies are also limited in that they have either used non-nationally representative data or relied on datasets that are more than ten years old. Moreover, few studies have examined whether the association between physical activity and subjective

memory decline differs across sex groups. Addressing these gaps using large, population-based data may clarify whether the protective association observed in selective samples holds in the broader U.S. population and across demographic subgroups, while helping identify groups most likely to benefit from early preventive interventions. Therefore, this study examined the association between physical activity and self-reported memory decline among U.S. adults aged 50 years and older using 2023 Behavioral Risk Factor Surveillance System (BRFSS) data, and further assessed whether this association differed by sex. We hypothesized that higher physical activity would be associated with lower odds of self-reported memory decline, with a weaker protective association among women than men, as prior studies have suggested that females experience more rapid progression of cognitive impairment once diagnosed[21,22].

Methods

Data source and study design

This research utilized the 2023 Behavioral Risk Factor Surveillance System (BRFSS) dataset, which is a cross-sectional survey of the nation's premier system of health-related telephone surveys. The survey was designed to gather state-level information from U.S. adults on their lifestyle and behavioral risk factors, the prevalence of chronic conditions, and their access to preventive and public-health services. The goal of the proposed study is to exam what the association is between self-reported memory decline and physical activity, and to assess whether sex may act as a modifier in their relation. Over 400k phone interviews were taken in 2023 in the US on the population(>=18 years), with a median response rate of 44.7%. For this analysis, we restricted the sample to non-institutionalized adults aged 50 years and above, English or Spanish speaking, residing in U.S. states/territories . After excluding respondents who answered "don't know," "refused," or had missing values on key variables, the final analytic sample was n= 35,366, representing a weighted population of 15,258,727 U.S. adults aged 50 years and older.

Covariates

The main outcome of interest is self-reported memory decline based on the Cognitive Decline Module question: "During the past 12 months, have you experienced difficulties with thinking or memory that are happening more often or are getting worse?" In BRFSS, responses were coded as 1 = Yes, 2 = No.

The main exposure was self-reported physical activity level, defined using the BRFSS calculated variable _PACAT3, which classifies respondents into four categories according to weekly minutes of aerobic physical activity or vigorous equivalent activity: inactive (no physical activity), insufficiently active (1-149 minutes/week), active (150-300 minutes/week), and highly active (>300 minutes/week).

Additionally, sex was examined as a potential effect modifier to assess whether the relationship betweenphysical activity and self-reported memory decline differs between men and women. In BRFSS, sex was coded as 1 = male and 2 = female.

Other variables examined in this study include educational level(never attended school or only kindergarten/ Elementary/ Some high school/ High school graduate/ Some college or technical school/ College graduate); (Ever told) have depressive disorder(yes/no); smoking frequency (Everyday/Someday/Not at all); Income level(Less than $15,000/ $15,000 to < $25,000/ $25,000 to < $35,000/ $35,000 to < $50,000/ $50,000 to < $100,000/ $100,000 to < $200,000/ $200,000 or more).

Statistical Analysis

Descriptive statistics were used to summarize participant characteristics overall and by self-reported memory decline status. For categorical variables, weighted frequencies and weighted percentages were reported; for any continuous variables, weighted means and standard deviations were summarized. In the bivariate table categorized by self-reported memory decline status (yes/no), p-values for categorical variables were obtained using design-adjusted survey chi-square tests, and p-values for any continuous variables were obtained using survey-weighted t-tests. To explore associations among categorical variables and inform covariate assessment, an unweighted correlation heatmap was constructed using Cramér's V. The heatmap included self-reported memory decline, physical activity, sex, age group, education, income, smoking status, and depressive disorder.

The primary analysis used survey-weighted logistic regression to estimate the association between physical activity and self-reported memory decline. Self-reported memory decline was

modeled as a binary outcome, with “No” as the reference category. Physical activity was modeled as a categorical exposure with “Inactive” as the reference group, and sex was coded with “Male” as the reference category. Odds ratios (ORs), 95% confidence intervals (CIs), and p-values were reported.

Three survey-weighted logistic regression models were fit. Model 1 was a baseline model including physical activity only. Model 2 was a multivariable-adjusted model including physical activity, sex, age group, education, income, smoking status, and depressive disorder. Model 3 further included interaction terms between physical activity and sex to assess whether the association between physical activity and self-reported memory decline differed by sex. In addition, two sex-stratified survey-weighted logistic regression models were fit separately for females and males, these models included physical activity, age group, education, income, smoking status, and depressive disorder and were used to estimate sex-specific odds ratios.

For all regression models, p-values were obtained from survey-weighted Wald tests of individual regression coefficients estimated using *svyglm* in R, which accounts for the complex survey design in the variance estimation. All statistical tests were two-sided, and statistical significance was assessed at an alpha level of 0.05.

Results

A total of 35,366 respondents met the analytic inclusion criteria, representing 15,258,727 weighted U.S. adults aged 50 years and older. In the weighted sample, 54% were male and 46% were female. The largest age group was 60-64 years (19%), followed by 65-69 years (16%), 50-54 years (15%), and 70-74 years (15%). With respect to physical activity, 44% of respondents were classified as highly active, 41% as inactive, 9.6% as active, and 5.3% as insufficiently active. Overall, 22% reported depressive disorder, and 21% reported self-reported memory decline (Table 1).

In bivariate analyses, self-reported memory decline differed significantly by physical activity level, sex, age group, education, income, and depressive disorder, but not by smoking status. Respondents with self-reported memory decline were more likely to be inactive and less likely to be highly active than those without memory decline (51% vs. 38% inactive, 36% vs. 46% highly active, $p<0.001$). A modest but statistically significant difference was also observed by sex, with females comprising a slightly larger proportion of the memory decline group than of the non-

decline group (49% vs. 45%, p=0.029). Age group distributions also differed by memory decline status (p=0.006), although the pattern was not strictly monotonic across categories. Respondents with self-reported memory decline were more likely to have lower education, lower income, and depressive disorder than those without memory decline (all p<0.001). Depressive disorder showed the strongest bivariate association, while smoking status was not significantly associated with memory decline (p=0.20).(Table 2)

The unweighted Cramér's V heatmap indicated generally weak to modest associations among variables. Depressive disorder showed the strongest crude association with self-reported memory decline (V=0.25), followed by income (V=0.14) and physical activity (V=0.10), and no substantial collinearity was apparent among predictors.(Figure 1)

Association between physical activity and self-reported memory decline: adjusted and sex-interaction models

In the baseline model including physical activity only, higher physical activity was associated with lower odds of self-reported memory decline relative to inactivity. Compared with inactive adults, the odds of memory decline were lower among insufficiently active adults (OR = 0.68, 95%CI:0.51-0.92), active adults (OR = 0.66, 95%CI: 0.52-0.82), and highly active adults (OR = 0.59, 95%CI:0.52-0.67), indicating a graded inverse association in the unadjusted analysis.

After adjustment for sex, age group, education, income, smoking status, and depressive disorder in Model 2, the association was attenuated. The strongest independent association remained for the highly active group, whose odds of self-reported memory decline were 24% lower than those of inactive adults (OR = 0.76, p < 0.001, 95%CI: 0.67-0.87). In contrast, the insufficiently active and active groups were no longer significantly different from the inactive group. Several covariates were also associated with the outcome in the adjusted model. Depressive disorder showed the strongest positive association with memory decline (OR = 3.88, p < 0.001,95%CI:3.42-4.41). Higher income categories were generally associated with lower odds of memory decline compared with the lowest income category, and some age groups, particularly 60-64 and 65-69 years, had lower odds relative to adults aged 50-54 years. In the adjusted model, female sex was not significantly associated with self-reported memory decline, though the result was near the conventional significance threshold. (OR = 0.89, p = 0.052, 95%CI:3.42-4.96).

Model 3 added interaction terms between physical activity and sex to assess whether the association between physical activity and self-reported memory decline differed between males and females. In this model, the main effect for highly active remained statistically significant (OR = 0.80, p = 0.014), whereas the main effects for insufficiently active and active remained non-significant. Most importantly, none of the interaction terms between physical activity and female sex were statistically significant: insufficiently active * female (OR = 1.50, p = 0.199), active * female (OR = 0.82, p = 0.391), and highly active * female (OR = 0.90, p = 0.421). These findings indicate that there was no clear evidence that sex modified the association between physical activity and self-reported memory decline. In other words, although the association between higher physical activity and lower odds of memory decline was observed, its magnitude did not differ significantly between males and females (Table 3).

Sex-stratified associations between physical activity and self-reported memory decline

In the female-stratified model, only the highly active group had significantly lower odds of self-reported memory decline compared with inactive females (OR = 0.73, p = 0.001). Neither the insufficiently active nor the active group differed significantly from the inactive group. Among female respondents, several age categories showed lower odds of memory decline relative to the reference age group (50-54), particularly ages 60-64, 65-69, and 75-79. Depressive disorder remained strongly associated with higher odds of memory decline (OR = 3.63, p < 0.001). A clear inverse income gradient was also observed, with several higher income categories associated with lower odds of memory decline.

In the male-stratified model, highly active males also had significantly lower odds of self-reported memory decline than inactive males (OR = 0.80, p = 0.019), whereas the insufficiently active and active groups were not significantly different from inactive males. Compared with the female model, the protective association for high physical activity appeared weaker in magnitude among males. Among male respondents, depressive disorder was also strongly associated with higher odds of memory decline (OR = 4.12, p < 0.001). In contrast to the female model, males aged 80 years and older had higher odds of memory decline than those aged 50-54, and lower educational attainment was associated with higher odds of memory decline in some categories. Male respondents who reported not smoking at all had modestly higher odds of memory decline than those who smoked every day. (Table 4)

Overall, the sex-stratified models suggest that the inverse association between high physical activity and self-reported memory decline was present in both females and males, with a stronger association among females. However, only the highly active category was consistently significant in both strata, while intermediate activity levels were not. These findings are broadly consistent with the interaction model, which did not provide evidence of statistically significant effect modification by sex.

Discussion

This study found that higher physical activity was associated with lower odds of self-reported memory decline among U.S. adults aged 50 years and older. The inverse association was most consistent among adults classified as highly active, even after adjustment for demographic, socioeconomic, and health-related factors.

To examine whether this protective association differed by sex, analyses were conducted separately for women and men. Among highly active women, the odds of self-reported memory decline were 27% lower relative to inactive women. In highly active men, the odds were 20% lower relative to inactive men. Both associations were statistically significant, indicating that the protective pattern of physical activity on subjective cognitive health was present and reliable in both sexes.

While the sex-stratified estimates suggest a numerically larger protective association in women, differences in point estimates across subgroups do not alone confirm a genuine population level distinction. Further test of whether sex modified the physical activity with self-reported memory decline association was not statistically significant (OR = 0.90, p = 0.42), indicating that the observed numerical difference between women and men was within the range of chance variation. There is therefore insufficient statistical evidence to conclude that the protective association of physical activity on memory decline differs meaningfully by sex.

These findings suggest that physical activity is an important approach correlate of subjective cognitive health in later life, with its protective benefits extending across both sexes. The findings only partially supported the original hypothesis. Although higher physical activity was associated with lower odds of self-reported memory decline as anticipated, the hypothesized weaker protective association among women was not observed. The sex-stratified estimates

were descriptively consistent with a somewhat stronger association in women. However, the absence of a statistically significant interaction indicates this pattern cannot be interpreted as evidence of a true sex difference. Overall, physical activity appears to be a broadly protective behavioral factor for cognitive health in this population, with benefits that are not meaningfully contingent on sex.

These results are consistent with prior literature indicating that physical activity supports cognitive health and may reduce dementia risk. They also extend previous work by focusing on self-reported memory decline as a potentially important early marker along the trajectory of cognitive impairment. The present findings suggest that physical activity may be relevant not only to later stage cognitive outcomes, but also to earlier perceived changes in memory that precede clinically recognized impairment. From public health perspectives, interventions targeting physical activity are likely to be most effective when initiated before substantial cognitive decline has already occurred.

The findings also point to the broader social and mental health context in which cognitive aging occurs. In the adjusted and stratified analyses, depressive disorder showed a strong and consistent association with memory decline, while socioeconomic indicators, particularly income, were also closely related to the outcome. This suggests that self-reported memory decline should not be considered in isolation from these broader contextual factors. From a population health standpoint, efforts to preserve cognitive health in older adults may benefit from approaches that pair physical activity promotion with depression screening in routine primary care encounters, and that prioritize targeted outreach to older adults in lower income communities where both physical inactivity and socioeconomic disadvantage converge.

The absence of a statistically significant sex difference is also informative. Although the introduction to this thesis outlined theoretical reasons to anticipate sex-differentiated effects, including historical differences in cognitive reserve accumulation through education, occupational attainment, and caregiving role, no such difference was detected in the present sample. One plausible interpretation is that within this cohort of adults aged 50 and older, who came of age during a period of more pronounced sex inequality in educational and occupational opportunity, the resulting similarity in lifetime cognitive reserve accumulation by sex may have limited the degree to which the protective effect of current physical activity would diverge between women and men. Consistent with this interpretation, the sex-stratified estimates were broadly comparable, and the magnitude of association varied only modestly across subgroups.

In practical terms, this finding suggests that physical activity promotion may serve as a broadly applicable strategy for cognitive health in the current older adult population.

This study has several public health implications. First, it supports physical activity as a practical, low-cost, and scalable component of strategies aimed at reducing the burden of cognitive decline and dementia in aging populations, particularly given that physical activity is already central to chronic disease prevention guidelines. Second, the use of self-reported memory decline as the outcome highlights the importance of recognizing early cognitive concerns in community settings. Public health systems and primary care settings may be able to use such early signals, together with behavioral risk assessment, to identify older adults who could benefit from preventive support before impairment develops. In addition to that, the strong association between memory decline and depressive disorder suggests that cognitive health promotion should be approached in an integrated way.

Limitations

This study has several limitations related to its cross-sectional design and data collection methodology. First, the Behavioral Risk Factor Surveillance System (BRFSS) relies on self-reported information, which may introduce recall bias or social desirability bias, particularly for sensitive topics such as memory decline, physical activity, or depressive symptoms. Because exposures and outcomes are measured simultaneously, causal relationships cannot be inferred. Also, non-response and selection bias may also affect the representativeness of the sample, as the median response rate of 44.7% suggests that individuals with poorer health or cognitive function might be underrepresented. In addition, residual confounding may remain due to unmeasured factors such as diet quality or genetic predisposition not captured in the dataset. Despite these limitations, the large sample size and standardized survey methods enhance the reliability of population-level estimates and support the study's public health relevance.

References

1. Jia L, Du Y, Chu L, Zhang Z, Li F, Lyu D. Prevalence, risk factors, and management of dementia and mild cognitive impairment in adults aged 60 years or older in China: a cross-sectional study. *Lancet Public Health*. Dec 2020;5(12):e661-e671. doi:10.1016/s2468-2667(20)30185-7
2. Dhana K, Beck T, Desai P, Wilson RS, Evans DA, Rajan KB. Prevalence of Alzheimer's disease dementia in the 50 US states and 3142 counties: A population estimate using the 2020 bridged-race postcensal from the National Center for Health Statistics. *Alzheimers Dement*. Oct 2023;19(10):4388-4395. doi:10.1002/alz.13081
3. Lee J, Meijer E, Langa KM, et al. Prevalence of dementia in India: National and state estimates from a nationwide study. *Alzheimers Dement*. Jul 2023;19(7):2898-2912. doi:10.1002/alz.12928
4. Juarez-Cedillo T, Gonzelez-Figueroa E, Gutierez-Gutierez L, et al. Prevalence of Dementia and Main Subtypes in Mexico: The Study on Aging and Dementia in Mexico (SADEM). *J Alzheimers Dis*. 2022;89(3):931-941. doi:10.3233/jad-220012
5. Numbers K, Lam BCP, Crawford JD, Kochan NA, Sachdev PS, Brodaty H. Longitudinal changes in participant and informant reports of subjective cognitive complaints are associated with dementia risk. *Front Aging Neurosci*. 2023;15:1044807. doi:10.3389/fnagi.2023.1044807
6. Numbers K, Crawford JD, Kochan NA, Draper B, Sachdev PS, Brodaty H. Participant and informant memory-specific cognitive complaints predict future decline and incident dementia: Findings from the Sydney Memory and Ageing Study. *PLoS One*. 2020;15(5):e0232961. doi:10.1371/journal.pone.0232961
7. Tsutsumimoto K, Makizako H, Doi T, et al. Subjective Memory Complaints are Associated with Incident Dementia in Cognitively Intact Older People, but Not in Those with Cognitive Impairment: A 24-Month Prospective Cohort Study. *Am J Geriatr Psychiatry*. Jun 2017;25(6):607-616. doi:10.1016/j.jagp.2016.12.008
8. Olmastroni E, Molari G, De Beni N, et al. Statin use and risk of dementia or Alzheimer's disease: a systematic review and meta-analysis of observational studies. *Eur J Prev Cardiol*. May 5 2022;29(5):804-814. doi:10.1093/eurjpc/zwab208
9. Fong TG, Inouye SK. The inter-relationship between delirium and dementia: the importance of delirium prevention. *Nat Rev Neurol*. Oct 2022;18(10):579-596. doi:10.1038/s41582-022-00698-7
10. Cabrera C, Vicens P, Torrente M. Modifiable Risk Factors for Dementia: The Role of Gut Microbiota. *Curr Alzheimer Res*. 2021;18(13):993-1009. doi:10.2174/1567205018666211215152411
11. Schippinger WM, Pichler G. [Prevention of dementia]. *Nervenarzt*. Mar 2024;95(3):283-290. Prävention demenzieller Erkrankungen. doi:10.1007/s00115-024-01641-5
12. Bossers WJ, van der Woude LH, Boersma F, Hortobágyi T, Scherder EJ, van Heuvelen MJ. A 9-Week Aerobic and Strength Training Program Improves Cognitive and Motor

Function in Patients with Dementia: A Randomized, Controlled Trial. *Am J Geriatr Psychiatry*. Nov 2015;23(11):1106-16. doi:10.1016/j.jagp.2014.12.191
13. Nuzum H, Stickel A, Corona M, Zeller M, Melrose RJ, Wilkins SS. Potential Benefits of Physical Activity in MCI and Dementia. *Behav Neurol*. 2020;2020:7807856. doi:10.1155/2020/7807856
14. Yu F, Lin FV, Salisbury DL, et al. Efficacy and mechanisms of combined aerobic exercise and cognitive training in mild cognitive impairment: study protocol of the ACT trial. *Trials*. Dec 22 2018;19(1):700. doi:10.1186/s13063-018-3054-0
15. Karssemeijer EGA, Aaronson JA, Bossers WJR, Donders R, Olde Rikkert MGM, Kessels RPC. The quest for synergy between physical exercise and cognitive stimulation via exergaming in people with dementia: a randomized controlled trial. *Alzheimers Res Ther*. Jan 5 2019;11(1):3. doi:10.1186/s13195-018-0454-z
16. 2024 Alzheimer's disease facts and figures. *Alzheimers Dement*. May 2024;20(5):3708-3821. doi:10.1002/alz.13809
17. Mielke MM. Sex and Gender Differences in Alzheimer's Disease Dementia. *Psychiatr Times*. Nov 2018;35(11):14-17.
18. Edland SD, Rocca WA, Petersen RC, Cha RH, Kokmen E. Dementia and Alzheimer disease incidence rates do not vary by sex in Rochester, Minn. *Arch Neurol*. Oct 2002;59(10):1589-93. doi:10.1001/archneur.59.10.1589
19. Gong J, Harris K, Lipnicki DM, et al. Sex differences in dementia risk and risk factors: Individual-participant data analysis using 21 cohorts across six continents from the COSMIC consortium. *Alzheimer's & Dementia*. 2023;19(8):3365-3378. doi:https://doi.org/10.1002/alz.12962
20. Seshadri S, Wolf PA, Beiser A, et al. Lifetime risk of dementia and Alzheimer's disease. The impact of mortality on risk estimates in the Framingham Study. *Neurology*. Dec 1997;49(6):1498-504. doi:10.1212/wnl.49.6.1498
21. Lin KA, Choudhury KR, Rathakrishnan BG, Marks DM, Petrella JR, Doraiswamy PM. Marked gender differences in progression of mild cognitive impairment over 8 years. *Alzheimers Dement (N Y)*. Sep 1 2015;1(2):103-110. doi:10.1016/j.trci.2015.07.001
22. Li A, Klinger HM, Seto M, et al. Elevated temporal tau PET predicts faster cognitive decline in women than men: A meta-analysis. *Alzheimers Dement*. Feb 2026;22(2):e71031. doi:10.1002/alz.71031
23. Barnes LL, Wilson RS, Bienias JL, Schneider JA, Evans DA, Bennett DA. Sex differences in the clinical manifestations of Alzheimer disease pathology. *Arch Gen Psychiatry*. Jun 2005;62(6):685-91. doi:10.1001/archpsyc.62.6.685
24. Giedd JN, Raznahan A, Mills KL, Lenroot RK. Review: magnetic resonance imaging of male/female differences in human adolescent brain anatomy. *Biol Sex Differ*. Aug 21 2012;3(1):19. doi:10.1186/2042-6410-3-19
25. Bretsky PM, Buckwalter JG, Seeman TE, et al. Evidence for an interaction between apolipoprotein E genotype, gender, and Alzheimer disease. *Alzheimer Dis Assoc Disord*. Oct-Dec 1999;13(4):216-21. doi:10.1097/00002093-199910000-00007
26. Rocca WA, Grossardt BR, Shuster LT. Oophorectomy, menopause, estrogen treatment, and cognitive aging: clinical evidence for a window of opportunity. *Brain Res*. Mar 16 2011;1379:188-98. doi:10.1016/j.brainres.2010.10.031

27. Simfukwe C, An SSA, Youn YC. Mechanisms and Impact of Cognitive Reserve in Normal Aging and Alzheimer's Disease. *Diagnostics (Basel)*. Dec 2 2025;15(23)doi:10.3390/diagnostics15233068
28. Stern Y. Cognitive reserve in ageing and Alzheimer's disease. *Lancet Neurol*. Nov 2012;11(11):1006-12. doi:10.1016/s1474-4422(12)70191-6
29. Schneider AL, Sharrett AR, Patel MD, et al. Education and cognitive change over 15 years: the atherosclerosis risk in communities study. *J Am Geriatr Soc*. Oct 2012;60(10):1847-53. doi:10.1111/j.1532-5415.2012.04164.x
30. Bureau USC. Census Bureau Releases New Educational Attainment Data. Accessed April 27, 2026, https://www.census.gov/newsroom/press-releases/2023/educational-attainment-data.html
31. Bellows-Riecken KH, Rhodes RE. A birth of inactivity? A review of physical activity and parenthood. *Prev Med*. Feb 2008;46(2):99-110. doi:10.1016/j.ypmed.2007.08.003
32. Milkie MAS, Liana C.; Nomaguchi, Kei; Yan, Hope Xu. Who's Doing the Housework and Childcare in America Now? Differential Convergence in Twenty-First-Century Gender Gaps in Home Tasks. *Socius*. 2025;11doi:10.1177/23780231251314667
33. Watson KB, Carlson SA, Gunn JP, et al. Physical Inactivity Among Adults Aged 50 Years and Older - United States, 2014. *MMWR Morb Mortal Wkly Rep*. Sep 16 2016;65(36):954-8. doi:10.15585/mmwr.mm6536a3
34. Lee PL. The relationship between memory complaints, activity and perceived health status. *Scand J Psychol*. Apr 2014;55(2):136-41. doi:10.1111/sjop.12107
35. McEwen SC, Siddarth P, Rahi B, et al. Simultaneous Aerobic Exercise and Memory Training Program in Older Adults with Subjective Memory Impairments. *J Alzheimers Dis*. 2018;62(2):795-806. doi:10.3233/jad-170846
36. de Souto Barreto P, Andrieu S, Rolland Y, Vellas B. Physical activity domains and cognitive function over three years in older adults with subjective memory complaints: Secondary analysis from the MAPT trial. *J Sci Med Sport*. Jan 2018;21(1):52-57. doi:10.1016/j.jsams.2017.07.019

Appendix

Table 1. Sociodemographic and Health Characteristics of U.S. Adults Aged 50 Years and Older, Behavioral Risk Factor Surveillance System (BRFSS), 2023, Weighted N = 15,258,727

Characteristic	N=15,258,727(%)
Sex	
Male	8,235,685 (54%)
Female	7,023,042 (46%)
Age(years)	
50-54	2,251,853 (15%)
55-59	2,162,951 (14%)
60-64	2,951,853 (19%)
65-69	2,478,136 (16%)
70-74	2,282,814 (15%)
75-79	1,519,890 (10.0%)
80+	1,611,228 (11%)
Physical Activity	
Inactive	6,249,893 (41%)
Insufficiently active	815,592 (5.3%)
Active	1,460,657 (9.6%)
Highly active	6,732,585 (44%)
Education	
Never attended/Kindergarten	27,269 (0.2%)
Grades 1–8	545,826 (3.6%)
Grades 9–11	1,337,829 (8.8%)
Grade 12 or GED	4,616,941 (30%)
College 1–3 years	5,202,617 (34%)
College 4+ years	3,528,244 (23%)
Income	
<$10k	541,541 (3.5%)
$10k–<$15k	611,752 (4.0%)
$15k–<$20k	749,503 (4.9%)
$20k–<$25k	1,109,660 (7.3%)
$25k–<$35k	1,938,433 (13%)
$35k–<$50k	2,374,065 (16%)
$50k–<$75k	2,715,058 (18%)
$75k–<$100k	1,898,156 (12%)
$100k–<$150k	1,872,037 (12%)
$150k–<$200k	737,870 (4.8%)
≥$200k	710,652 (4.7%)

Smoke	
Every day	3,276,367 (21%)
Some days	1,075,467 (7.0%)
Not at all	10,906,893 (71%)
Depression	3,303,269 (22%)
mem_decline	3,272,604 (21%)

Table 2. Characteristics of U.S. Adults Aged 50 Years and Older by Self-Reported Memory Decline Status, Behavioral Risk Factor Surveillance System (BRFSS 2023) Weighted N = 15,258,727

Variable	Self-reported Memory Decline		
	No	Yes	p-value[1]
	N=11,986,124(%)	N = 3,272,604(%)	
Physical Activities			<0.001
Inactive	4,594,275 (38%)	1,655,618 (51%)	
Insufficiently active	654,631 (5.5%)	160,961 (4.9%)	
Active	1,181,816 (9.9%)	278,841 (8.5%)	
Highly active	5,555,403 (46%)	1,177,182 (36%)	
Sex			0.029
Male	6,551,445 (55%)	1,684,240 (51%)	
Female	5,434,679 (45%)	1,588,363 (49%)	
Age(years)			0.006
50–54	1,696,627 (14%)	555,226 (17%)	
55–59	1,670,809 (14%)	492,142 (15%)	
60–64	2,331,877 (19%)	619,977 (19%)	
65–69	2,024,006 (17%)	454,130 (14%)	
70–74	1,802,193 (15%)	480,622 (15%)	
75–79	1,226,763 (10%)	293,127 (9.0%)	
80+	1,233,849 (10%)	377,380 (12%)	
Education			<0.001
Never attended/Kindergarten	21,981 (0.2%)	5,288 (0.2%)	
Grades 1–8	352,677 (2.9%)	193,149 (5.9%)	
Grades 9–11	994,834 (8.3%)	342,995 (10%)	
Grade 12 or GED	3,574,682 (30%)	1,042,259 (32%)	
College 1–3 years	4,137,855 (35%)	1,064,763 (33%)	
College 4+ years	2,904,096 (24%)	624,149 (19%)	

Income			<0.001
<$10k	346,419 (2.9%)	195,122 (6.0%)	
$10k–<$15k	417,785 (3.5%)	193,967 (5.9%)	
$15k–<$20k	533,037 (4.4%)	216,466 (6.6%)	
$20k–<$25k	824,483 (6.9%)	285,178 (8.7%)	
$25k–<$35k	1,455,218 (12%)	483,214 (15%)	
$35k–<$50k	1,837,123 (15%)	536,942 (16%)	
$50k–<$75k	2,183,685 (18%)	531,374 (16%)	
$75k–<$100k	1,540,790 (13%)	357,366 (11%)	
$100k–<$150k	1,603,300 (13%)	268,737 (8.2%)	
$150k–<$200k	616,198 (5.1%)	121,673 (3.7%)	
≥$200k	628,086 (5.2%)	82,566 (2.5%)	
Smoke			0.2
Every day	2,520,062 (21%)	756,306 (23%)	
Some days	835,775 (7.0%)	239,692 (7.3%)	
Not at all	8,630,287 (72%)	2,276,606 (70%)	
Depression	1,903,344 (16%)	1,399,925 (43%)	<0.001

[1]p-value calculated by chi-square test

Table 3. Crude and adjusted odds ratios for the association between physical activity and self-reported memory decline among U.S. adults aged 50 years and older, Behavioral Risk Factor Surveillance System (BRFSS), United States, 2023

	Crude OR (95%CI)	Full Model Adjusted OR (95% CI)	Full Model with Interaction term PA*sex OR (95%CI)
Physical Activities			
Inactive	Ref	Ref	Ref
Insufficiently active	0.682 (0.507,0.919)	0.807 (0.6,1.086)	0.642 (0.385,1.072)
Active	0.655 (0.524,0.818)	0.904 (0.714,1.146)	0.992 (0.696,1.413)
Highly active	0.588 (0.520,0.665)	0.764 (0.668,0.872)	0.801 (0.671,0.957)
Sex			
Male		Ref	Ref
Female		0.887 (0.786,1.001)	0.921 (0.770,1.102)
Age(years)			
50–54		Ref	Ref

55–59	0.873 (0.698,1.091)	0.876 (0.701,1.094)
60–64	0.803 (0.649,0.994)	0.806 (0.651,0.998)
65–69	0.671 (0.540,0.835)	0.671 (0.540,0.835)
70–74	0.854 (0.675,1.080)	0.853 (0.675,1.078)
75–79	0.822 (0.655,1.033)	0.823 (0.655,1.034)
80+	1.116 (0.877,1.419)	1.117 (0.878,1.421)
Education		
Never attended/Kindergarten	Ref	Ref
Grades 1–8	2.563 (0.5,13.155)	2.550 (0.502,12.966)
Grades 9–11	1.559 (0.318,7.650)	1.543 (0.317,7.500)
Grade 12 or GED	1.560 (0.321,7.578)	1.549 (0.322,7.457)
College 1–3 years	1.445 (0.297,7.043)	1.431 (0.296,6.908)
College 4+ years	1.427 (0.292,6.977)	1.416 (0.296,6.908)
Income		
<$10k	Ref	Ref
$10k–<$15k	0.798 (0.551,1.154)	0.795 (0.549,1.150)
$15k–<$20k	0.836 (0.570,1.226)	0.834 (0.569,1.222)
$20k–<$25k	0.659 (0.470,0.923)	0.658 (0.470,0.921)
$25k–<$35k	0.743 (0.535,1.031)	0.740 (0.533,1.027)
$35k–<$50k	0.671 (0.486,0.926)	0.670 (0.486,0.924)
$50k–<$75k	0.592 (0.422,0.831)	0.590 (0.421,0.828)
$75k–<$100k	0.578 (0.410,0.813)	0.576 (0.408,0.811)

$100k–<$150k	0.419 (0.296,0.593)	0.416 (0.294,0.589)
$150k–<$200k	0.463 (0.302,0.711)	0.461 (0.301,0.709)
≥$200k	0.363 (0.238,0.554)	0.360 (0.236,0.550)
Smoke		
Every day	Ref	Ref
Some days	0.983 (0.761,1.270)	0.987 (0.763,1.275)
Not at all	1.168 (1.004,1.360)	1.169 (1.005,1.360)
DepressionYes	3.879 (3.415,4.406)	3.874 (3.410,4.400)
Physical Activity*Sex		
Insufficiently active:Female		1.499 (0.809,2.777)
Active:Female		0.818 (0.516,1.295)
Highlyactive:Female		0.902 (0.701,1.160)

OR=Odds ratio

CI = Confidence Interval

Table 4. Sex-stratified adjusted odds ratios for the association between physical activity and self-reported memory decline among U.S. adults aged 50 years and older, Behavioral Risk Factor Surveillance System (BRFSS 2023), United States

	Stratified OR-Female (95% CI)	**Stratified OR-Male (95% CI)**
Physical Activities		
Inactive	Ref	Ref
Insufficiently active	0.942(0.659,1.348)	0.657(0.392,1.101)
Active	0.797(0.591,1.075)	1.003(0.707,1.424)
Highly active	0.727(0.599,0.882)	0.803(0.668,0.965)
Age(years)		
50–54	Ref	Ref
55–59	0.855(0.638,1.145)	0.878(0.626,1.232)
60–64	0.670(0.509,0.883)	0.952(0.688,1.317)
65–69	0.597(0.445,0.802)	0.738(0.539,1.011)
70–74	0.740(0.535,1.025)	0.959(0.685,1.342)

75–79	0.706(0.521,0.957)	0.931(0.666,1.303)
80+	0.767(0.560,1.052)	1.425(1.001,2.029)
Education		
Never attended/Kindergarten	Ref	Ref
Grades 1–8	0.008(0.001,0.122)	5.839(1.402,24.310)
Grades 9–11	0.004(0.000,0.059)	3.811(0.983,14.769)
Grade 12 or GED	0.004(0.000,0.057)	3.986(1.052,15.105)
College 1–3 years	0.004(0.000,0.056)	3.444(0.904,13.118)
College 4+ years	0.004(0.000,0.055)	3.387(0.880,13.033)
Income		
<$10k	Ref	Ref
$10k–<$15k	0.663(0.404,1.088)	1.044(0.616,1.770)
$15k–<$20k	0.656(0.398,1.080)	1.175(0.705,1.961)
$20k–<$25k	0.572(0.349,0.938)	0.836(0.543,1.285)
$25k–<$35k	0.650(0.403,1.049)	0.907(0.600,1.371)
$35k–<$50k	0.655(0.409,1.048)	0.746(0.501,1.111)
$50k–<$75k	0.525(0.310,0.889)	0.700(0.472,1.039)
$75k–<$100k	0.447(0.274,0.731)	0.760(0.489,1.181)
$100k–<$150k	0.447(0.264,0.755)	0.435(0.286,0.664)
$150k–<$200k	0.373(0.208,0.670)	0.593(0.339,1.039)
≥$200k	0.294(0.159,0.543)	0.458(0.268,0.784)
Smoke		
Every day	Ref	Ref
Some days	1.010(0.710,1.437)	0.960(0.653,1.412)
Not at all	1.071(0.876,1.311)	1.308(1.049,1.631)
DepressionYes	3.631(3.057,4.313)	4.120(3.422,4.961)

OR = Odds ratio

Figure 1. Heatmap of Pairwise Associations Among Study Variables Measured by Cramér's V, BRFSS 2023

